Heart Disease MLOps Assignment Report

This report documents the end-to-end MLOps workflow implemented for the heart disease classification task. The project packages a reproducible ML pipeline, tests, metrics, a containerized API, and deployment artifacts.

Project scope: UCI Heart Disease dataset, feature preprocessing, model comparison, MLflow tracking, FastAPI serving, Docker, Kubernetes, and monitoring outputs.

1. Objective and dataset

The project aims to predict heart disease risk from health features using a reproducible ML pipeline. The Cleveland Heart Disease dataset provides the training data and is handled in the data pipeline module.

2. Data acquisition and EDA

The data module downloads the feature set, validates the schema, converts the target to binary values, and saves a cleaned CSV. Exploratory analysis includes class-balance, histogram, and correlation checks.

3. Model development

Two classifiers were evaluated: Logistic Regression and Random Forest. The selection criterion was mean cross-validation ROC-AUC. The final metrics retained in artifacts/metrics.json show excellent discrimination.

4. Experiment tracking

MLflow logs parameters, model metadata, metrics, and plots for both candidate runs. The local tracking store is retained under mlruns and can be inspected with the MLflow UI.

5. Automated testing and CI/CD

The repository includes tests for data validation and API behavior, and a GitHub Actions workflow validates linting, tests, dataset creation, and model training automatically.

6. Containerization and deployment

The service is served via FastAPI and runs inside a Docker container. Kubernetes deployment manifests provide the deployment blueprint for local cluster use.

7. Monitoring and operational proof

The API exposes /metrics and logs prediction requests, making it suitable for lightweight monitoring and operational checks in a local or containerized environment.

8. Evidence and repository artifacts

The repository includes model artifacts, plots, tests, deployment manifests, and a screenshot folder containing generated evidence images.

Key metrics

Selected model: Logistic Regression

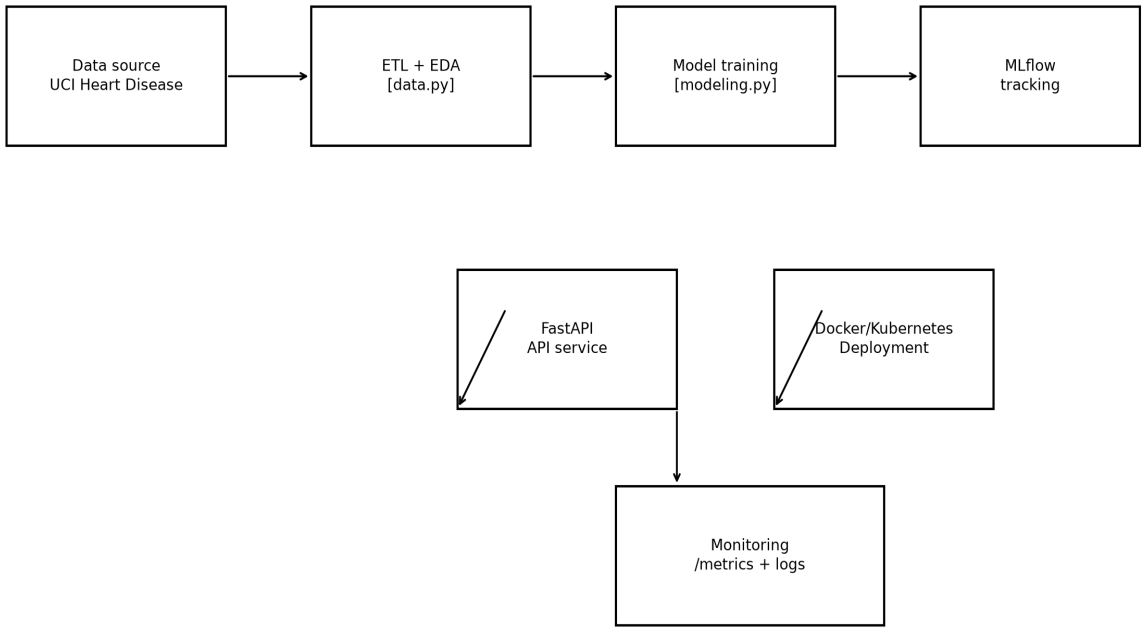
CV ROC-AUC: 0.9001

Test accuracy: 0.8525

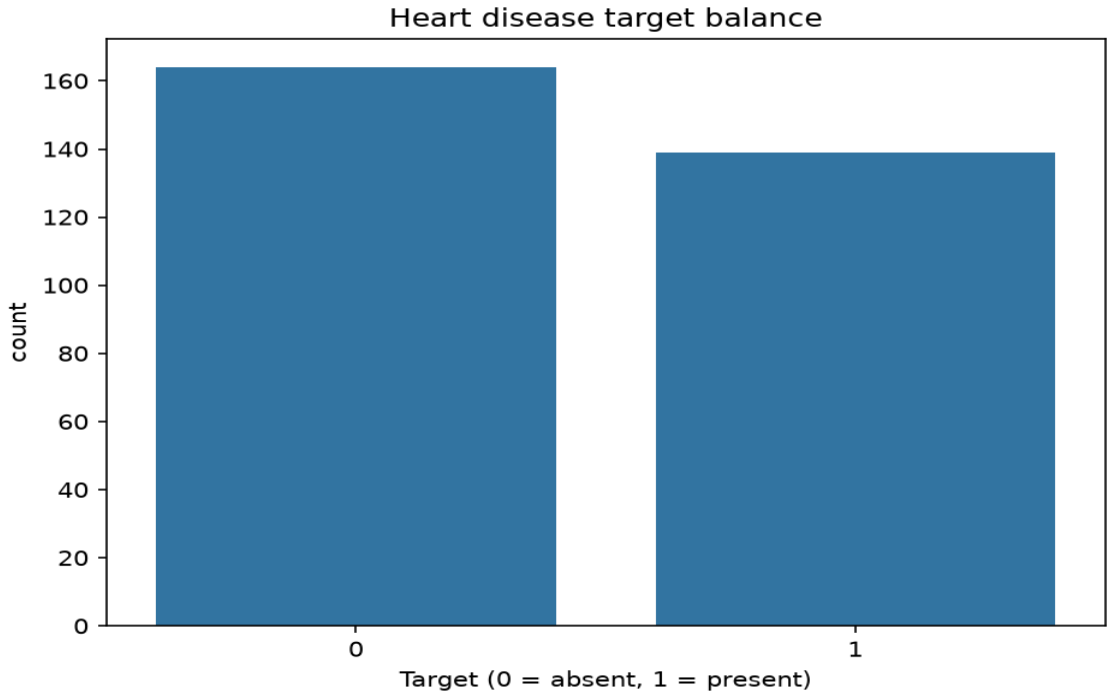
Test precision: 0.8065

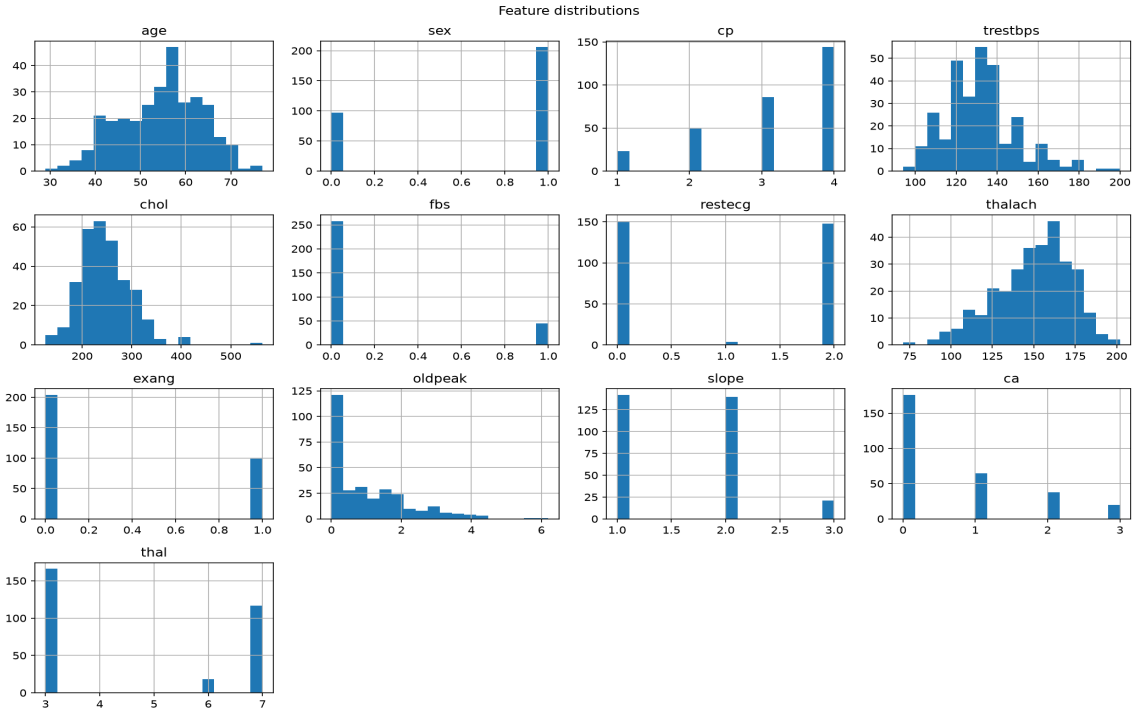
Test recall: 0.8929

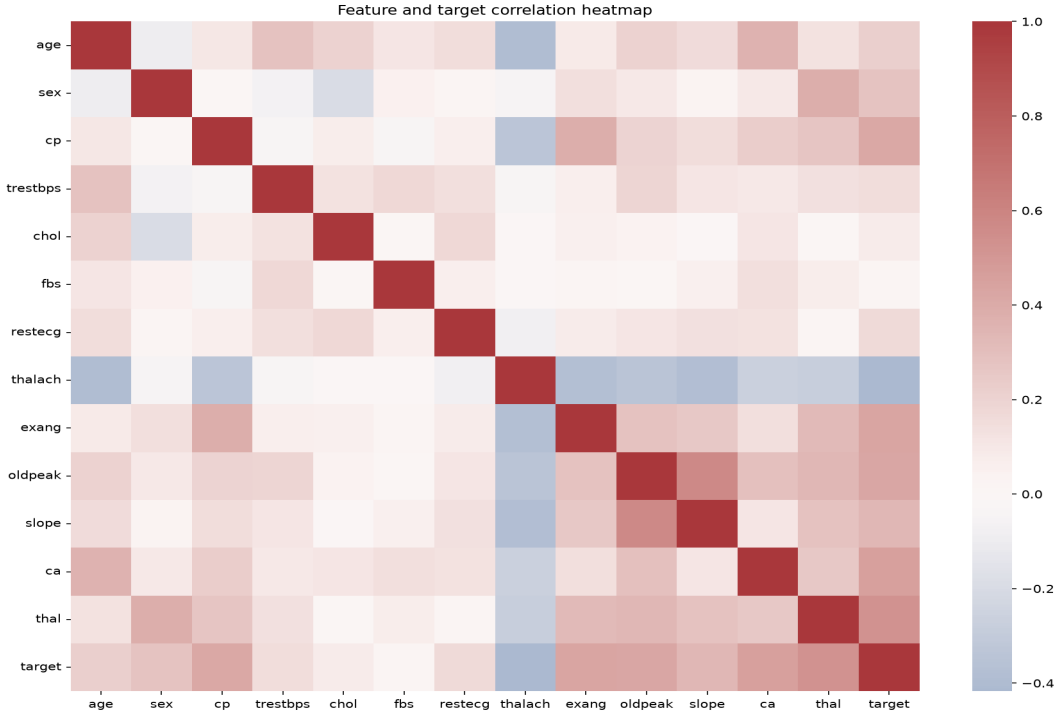
Test ROC-AUC: 0.9578



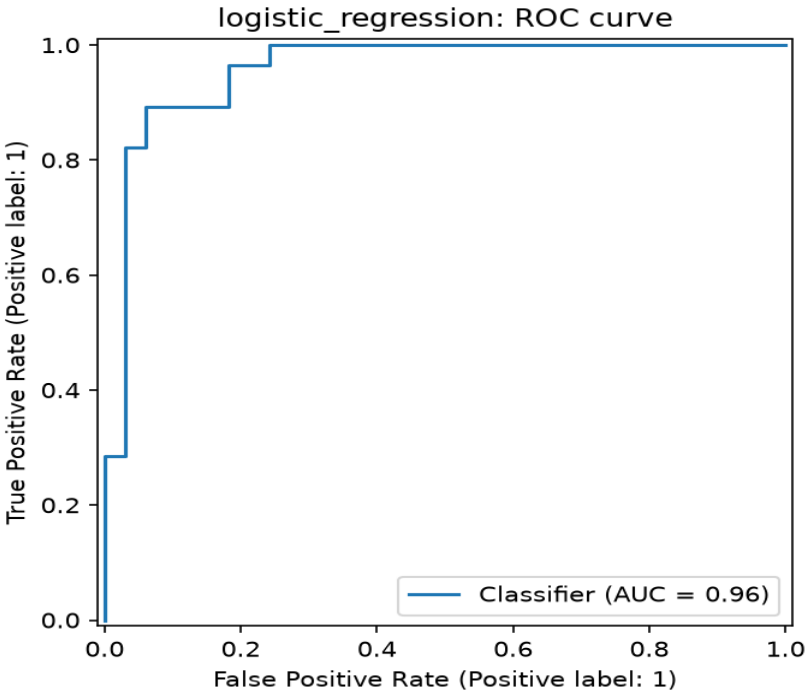
EDA evidence

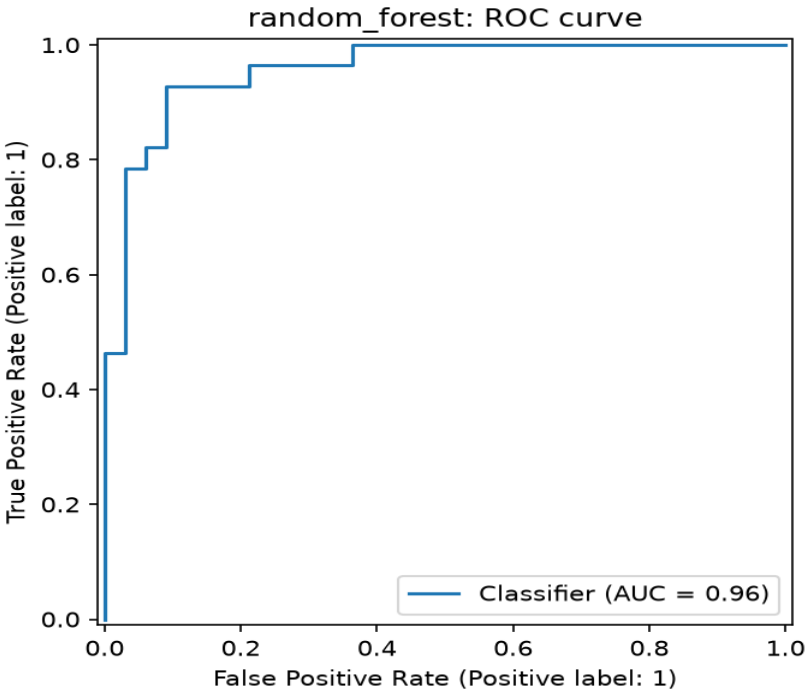





Feature and target correlation heatmap
age
sex
cp
trestbps
chol
fbs
restecg
thalach
exang
oldpeak
slope
ca
thal
target
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4

Model result evidence





The assignment video component remains for the student to record separately, while the rest of the technical and documentation requirements are included in the repository as required.